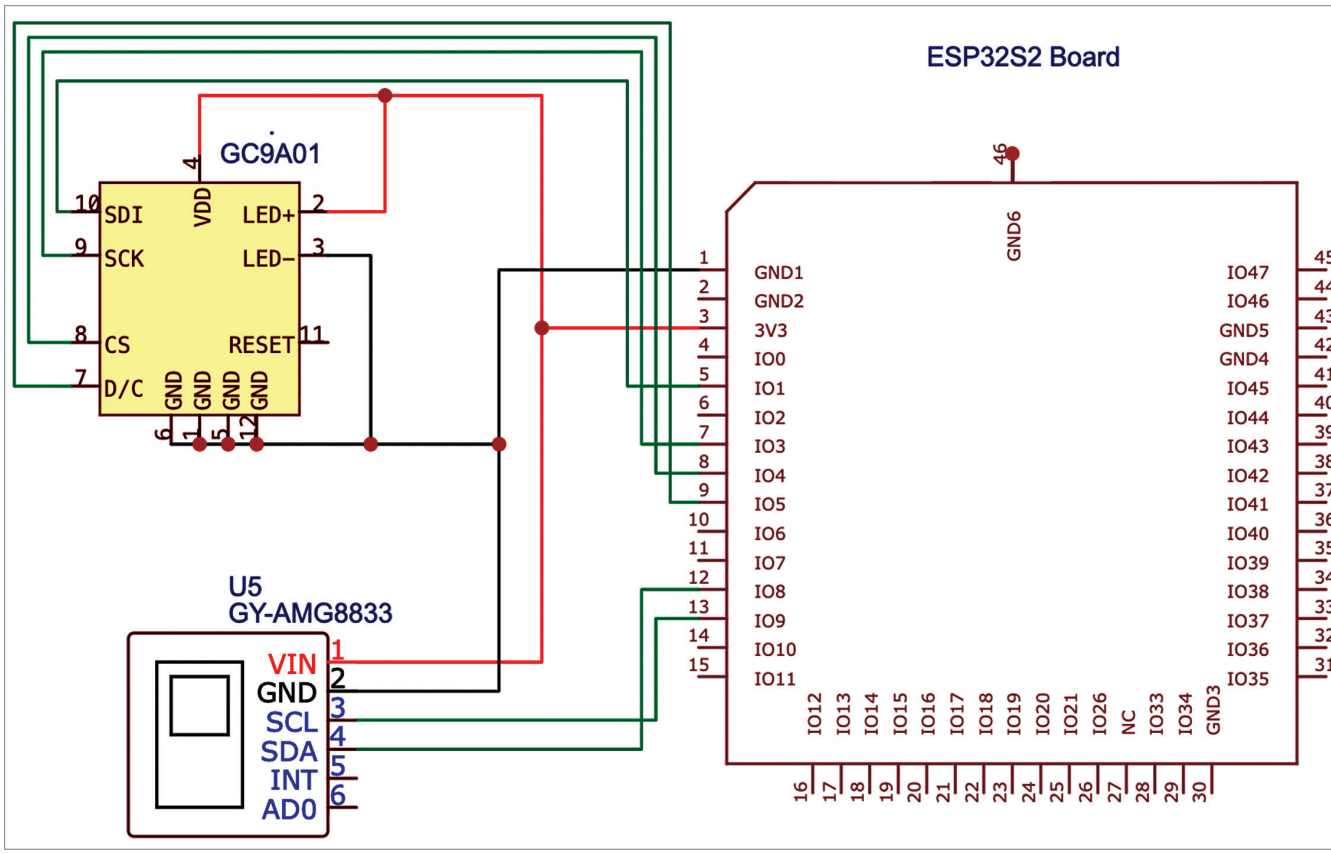


Fig. 3: Circuit of thermal camera using any ESP32S2 chip based board

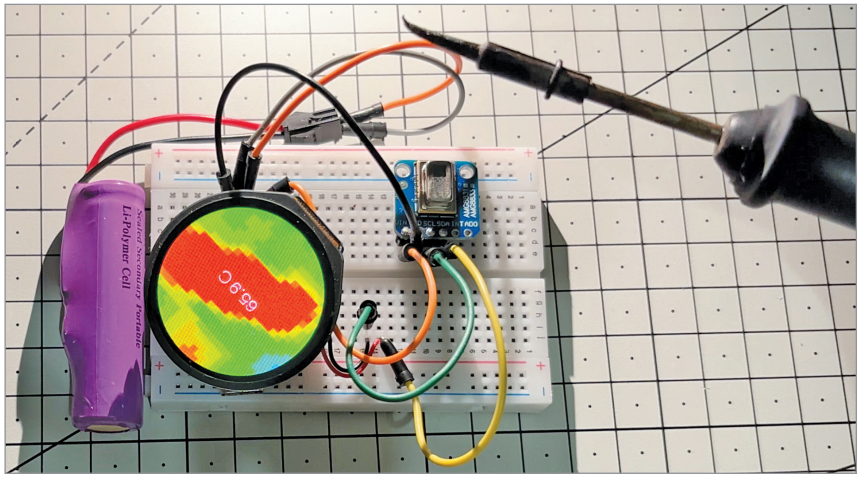


Fig. 4: Thermal camera on breadboard

GPIO pins can be assigned according to the circuit. For this system, the GC9A01 display driver should be selected and the pins configured accordingly.

During operation, the code continuously reads the 8 × 8 temperature data from the AMG8833, maps the readings to a cold-to-hot colour gradient (blue→yellow→red), and displays the resulting thermal image in real time. Fig. 6 shows a code snippet for mapping the thermal grid data to display colours.

Testing and troubleshooting

After uploading the code, the device can be powered using a 3.7V Li-Po battery or USB. The round display should light up and show a live thermal image within a few seconds. Placing a finger or hand in front of the AMG8833 sensor should produce a yellow-to-red heat region on the screen along with the centre temperature reading.

Common issues and solutions

1. Display remains blank. Check the SPI connections (CS, SCK, MOSI, and D/C) and verify that the correct GPIO pins are defined in the User_Setup.h file of TFT_eSPI. Also ensure that the GC9A01 driver is enabled.

2. No thermal image or fixed blue screen. Check the AMG8833 I²C connections (SDA and SCL) and verify that the Adafruit AMG88xx library is installed correctly.

3. Wrong or inverted colours. Adjust the MINTEMP and MAXTEMP values in the code. A typical range for human-body